



**Federal Aviation
Administration**

Initial En Route Qualification Training

Lesson 06 Recording Clearances and Control Information

Course 50148001

LESSON PLAN DATA SHEET

COURSE NAME: INITIAL EN ROUTE QUALIFICATION TRAINING
COURSE NUMBER: 50148001

LESSON TITLE: RECORDING CLEARANCES AND CONTROL INFORMATION

DURATION: 6+00 HOURS

DATE REVISED: 2025-02
VERSION: V.2025-02

REFERENCE(S): FAA ORDERS JO 7110.65, AIR TRAFFIC CONTROL, AND JO 7340.2, CONTRACTIONS

HANDOUT(S): stripmrk.f2k – STRIPMARKING EXERCISE STRIPS

**EXERCISE(S)/
ACTIVITY(S):** ACTIVITY 1: IDENTIFYING ABBREVIATIONS
ACTIVITY 2: IDENTIFYING CONTROL INFORMATION SYMBOLS
EXERCISE 1: RECORDING CLEARANCES AND CONTROL INFORMATION
EXERCISE 2: STRIPMARKING

**END-OF-LESSON
TEST:** YES

**PERFORMANCE
TEST:** NONE

MATERIALS: NONE

**OTHER PERTINENT
INFORMATION:**

DISCLAIMER

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INTRODUCTION

Initial En Route Qualification Training

Lesson 06 Recording Clearances and Control Information

V.2025-02
Presented by
FAA Academy
Air Traffic Division



Federal Aviation
Administration



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Correctly issuing and recording clearances and control information is essential to the safe and expeditious flow of traffic in a nonradar environment. Failure to properly record clearance information in a timely manner can lead to subsequent clearances which can result in a loss of separation or an aircraft accident. In this lesson, you will build on your knowledge of flight progress strips by learning control symbology and recording information in the correct spaces.

INTRODUCTION *(Continued)*

RECORDING CLEARANCES AND CONTROL INFORMATION						
CORRECT STRIPMARKING						
N991L	GLH 1840	55 18 -58-	110✓ ↑ 130	MEI	KGLH V74 MHZ V18 KMEI/1915	6621
BE9L/A T210 G210						
66						
117 02		MHZ				

INCORRECT STRIPMARKING						
N991L	GLH 1840	55 18 58	110✓ 130	MEI	KGLH V74 MHZ V18 KMEI/1915	6621
BE9L/A T210 G210						
66						
117 02		MHZ				

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Purpose

The material in this lesson will demonstrate correct stripmarking, which will help you maintain the current status of each aircraft in your sector.

INTRODUCTION *(Continued)*

Lesson Objectives

LESSON OBJECTIVES

- On an End-of-Lesson Test, and in accordance with FAA Order JO 7110.65 and JO 7340.2, you will identify:
 - Select stripmarking symbology
 - Select stripmarking abbreviations
 - Procedures for recording clearance and control information on flight strips

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CONTROL SYMBOLOGY

General Information

JO 7110.65,
pars. 2-3-9, 2-3-10

- ☉ Use authorized symbols or abbreviations for recording:
 - Clearances
 - Reports
 - Instructions
- ☉ You may use:
 - Plain language markings when necessary for clarification
 - Locally approved identifiers/abbreviations within your facility
- ☉ Use appropriate clearance symbol followed by a dash (-) and other information to show status of aircraft.

NOTE: The upcoming sections give more detailed information about the various abbreviations and symbols for use when marking strips.

CLEARANCE AND MISCELLANEOUS ABBREVIATIONS

Clearance Abbreviations

JO 7110.65,
par. 2-3-10,
table 2-3-11

☉ Clearance abbreviations include:

- **D** – Cleared to Depart from Fix
- **A** – Cleared to Destination Airport
- **F** – Cleared to Fix
- **H** – Cleared to Hold and Instructions Issued
- **V** – Cleared Over Fix
- **Z** – Tower Jurisdiction
- **PD** – Cleared to Climb/Descend at Pilot's Discretion

Cleared to Depart from Fix

JO 7110.65,
par. 2-3-10

D CLEARED TO DEPART FROM FIX					
N234		↑	HEZ	KJAN MHZ V245 KHEZ/0030	D-A ZHU
C310/A T180		1555/			
66	01	KJAN P1600	80		

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☉ Used in space 28 on originating IFR clearance strip **only**

- Departing from airport
- Airfile

Dash From – To

JO 7110.65,
par. 2-3-10

- (DASH) FROM – TO (ROUTE, TIME, ETC.)						
N975		T → NE TL 330/⇒V417	↑80	MLU	KVKS MLU V7 KELD/0037	D-A ZFW
C310/A T180		1555/				
66	01	KVKS P1600	80			

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☉ Follows “D” in space 28

CLEARANCE AND MISCELLANEOUS ABBREVIATIONS

(Continued)

Cleared to Destination Airport

JO 7110.65,
par. 2-3-10

A CLEARED TO DESTINATION AIRPORT						
N975		T → NE TL 330/⇒V417	↑80	MLU	KVKS MLU V7 KELD/0037 V417	D-A
C310/A T180						
66		1555/				
01		KVKS P1600		80		ZFW

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- ⊙ Follows “- (Dash)” in space 28

Cleared to Fix

JO 7110.65,
par. 2-3-10

F CLEARED TO FIX TO POSTED FIX						
UAL903	GLH 0022	26 00	110✓	IGB	KLIT GLH V278 IGB KGTR	F
DC91/A T450		26				
66		SQS				
03						

TO OTHER THAN POSTED FIX						
N21DC	MLU 0005	20 00	80✓	MHZ 0048	KDAL MLU V427 MHZ V18 KMEI/0110	F-MHZ
C210/A T190		20				
66		HATER				
01						

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- ⊙ Used in space 28
- ⊙ “F – (fix)” is used when clearance limit is other than the posted fix.
- ⊙ Fix need **not** be recorded if:
 - The aircraft is cleared to the posted fix
 - Holding instructions have been issued and recorded

CLEARANCE AND MISCELLANEOUS ABBREVIATIONS

(Continued)

Cleared to Hold and Instructions Issued

JO 7110.65,
par. 2-3-10,
4-6-1, 4-6-4



Phraseology Example

H CLEARED TO HOLD AND INSTRUCTIONS ISSUED						
(FIX) (LEG)						
UAL36 B721/A T420 66 02	GLH 1636	41 16 41 SQS	100✓	IGB	KDAL GLH V278 IGB KGTR	H 20 10 NE -086 1650
“UNITED THIRTY-SIX CLEARED TO SIDON ZERO EIGHT SIX RADIAL TWO ZERO MILE FIX. HOLD NORTHEAST ON THE ZERO EIGHT SIX RADIAL ONE ZERO MILE LEG. EXPECT FURTHER CLEARANCE ONE SIX FIVE ZERO.”						

- ⊙ Used in space 28
- ⊙ Holding instructions **must** include:
 - Direction from fix
 - Holding fix, if **not** posted fix
 - Upper portion of “H” indicates distance from the station to the fix
 - Radial, course, azimuth, or route on which aircraft will hold
 - Leg length, if other than standard
 - Indicated in lower portion of H and shown in:
 - Minutes, or
 - Miles if DME is used
 - Direction of turns
- ⊙ Additional information, such as holding airspeed or Expect Further Clearance (EFC) time, is sometimes included in holding instructions.

Continued on next page

CLEARANCE AND MISCELLANEOUS ABBREVIATIONS

(Continued)

Cleared to Hold and Instructions Issued (Cont'd)

JO 7110.65,
par. 2-3-10
4-6-1, 4-6-4



Phraseology Example

H CLEARED TO HOLD AND INSTRUCTIONS ISSUED (CONT'D)

N36LJ	GLH	40	110✓	KGWO	KGLH V278 SQS	H SW -256 LT
LJ25/A	1635	16		1647	KGWO/1645	
T420		39				
66		SQS				
02						

"LEAR THREE SIX LIMA JULIETT, CLEARED TO SIDON VORTAC. HOLD SOUTHWEST ON THE TWO FIVE SIX RADIAL, LEFT TURNS. NO DELAY EXPECTED."

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Continued on next page

CLEARANCE AND MISCELLANEOUS ABBREVIATIONS

(Continued)

Cleared Over Fix

JO 7110.65,
par. 2-3-10



Phraseology Example

V **CLEARED OVER FIX**

Cancels previously issued clearance limit and holding instructions.

UAL36	GLH	41	170✓	IGB	KDAL GLH V278 IGB KGTR	
B721/A T420	1636	16				
66		41				
02		SQS				

“UNITED THIRTY-SIX, CLEARED TO GOLDEN TRIANGLE AIRPORT VIA VICTOR TWO SEVENTY-EIGHT BIGBEE DIRECT.”

OR

“UNITED THIRTY-SIX, CLEARED TO GOLDEN TRIANGLE AIRPORT VIA LAST ROUTING CLEARED.”

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⦿ Used in space 28 to mark over:

- F – Clearance limit
- H – Holding instructions

CLEARANCE AND MISCELLANEOUS ABBREVIATIONS

(Continued)

Tower Jurisdiction

JO 7110.65,
par. 2-3-10

Z TOWER JURISDICTION							
DAL61	HEZ	01	10	130✓ ↓60	KJAN	KMSY../HEZ V245 MHZ KJAN	 26 SW
B721/A T420	0100	10					
66	01	MHZ		↓60			

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⊙ Used in space 28

Cleared to Climb/Descend at Pilot's Discretion

JO 7110.65,
par. 2-3-10

PD CLEARED TO CLIMB/DESCEND AT PILOT'S DISCRETION							
AAL48	GLH	52	22	200✓ ↓160 PD	MCB	KSTL GLH V74 MHZ V9 KMCB	ZHU
B721/A T420	2247	52					
66	02	MHZ					

“AMERICAN FORTY-EIGHT, DESCEND AT PILOT'S DISCRETION, MAINTAIN ONE SIX THOUSAND.”

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Phraseology
Example

⊙ Used in space 20

Miscellaneous Abbreviations

JO 7110.65,
par. 2-3-10 ;
JO 7340.2

⊙ Miscellaneous abbreviations include:

- TL/TR – Turn Left/Turn Right
- HFR – Hold for Release
- RLS – Release
- SYD – Release Subject Your Discretion
- RR/RL/RP/RX – Request for Altitude/Fix Reports

CLEARANCE AND MISCELLANEOUS ABBREVIATIONS

(Continued)

Turn Left/Turn Right

JO 7110.65,
par. 2-3-10
TBL2-3-12

TL/TR TURN LEFT / TURN RIGHT			
DAL338		T→N TL	↑
DC91A			
T420			
66		1313/	
01		KGWO P1315	

VV83578		T→RY18 TR	↑
P3/B			
T300			
66		1320/	
01		KGWO P1325	

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TL/TR TURN LEFT / TURN RIGHT (CONT'D)			
N3721K		T→N TL 300	↑
C310/A			
T160			
66		1927/	
01		KGWO P1930	

AAL63		T→S TR 250	↑
B722/A			
T450			
66		1913/	
01		KGWO P1915	

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☉ Used in space 15 with magnetic heading to be flown

- 001 through 360

CLEARANCE AND MISCELLANEOUS ABBREVIATIONS

(Continued)

Hold For Release

JO 7110.65,
par. 4-3-4;
JO 7340.2

HFR HOLD FOR RELEASE						
N234		T → SW TR 240 / ⇒ V11	↑90	MHZ	KGWO SQS V11 MHZ V18 KMEI/0100	D-A
C310/A T180		HFR				
66						
01		KGWO P1600		90		
15						

- ⊙ Used in space 15 in conjunction with departure clearances
- ⊙ Issued to FDU or tower

Release

JO 7340.2



Phraseology Example

RLS RELEASE						
N234		T → SW TL 150/ ⇒ V278	↑90	IGB	KGWO SQS V278 IGB KUBS/0035	D-A
C310/A T180		RLS 1 MIN<N456				
66		1558/				
01		KGWO P1600		90		
16						
N456		T → SW TR 240/ ⇒ V278	↑80	GLH	KGWO SQS V278 KGLH/0014	D-A
C310/A T180						
66		1557/				
01		KGWO P1600		80		
“TWIN CESSNA TWO THREE FOUR, RELEASED ONE MINUTE AFTER TWIN CESSNA FOUR FIVE SIX DEPARTS.”						
16						

- ⊙ Used in space 15 in conjunction with departure clearance
 - When aircraft can be released for departure
- ⊙ Issued to tower

Continued on next page

CLEARANCE AND MISCELLANEOUS ABBREVIATIONS

(Continued)

**Release
Subject Your
Discretion**
JO 7340.2

✈
**Phraseology
Example**

SYD RELEASE SUBJECT YOUR DISCRETION							
N456 C310/A T180 66 01		T → SW -SQS SYD/N234	↑ 80	GLH 70	KGWO SQS V278 KGLH/0015	D-A	
		1651/ KGWO P1651					
N234 C310/A T180 66 02	GLH 1638	50 16 50 1649 SQS	↓ 70✓	KGWO 1657	KGLH V278 SQS KGWO/1655 C 67 ↓ 70	VR VR 1645	

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- ⦿ Used in space 15
- ⦿ Issued to tower when using visual separation

CLEARANCE AND MISCELLANEOUS ABBREVIATIONS

(Continued)

Request for Altitude/Fix Reports

JO 7110.65,
par. 2-3-10;
JO 7340.2



Phraseology Example

RL REPORT LEAVING							
N31B	IGB	35	120↓40	HEZ	KSTF IGB V245 KHEZ/1600		
BE55/A T190	1501	15	RL 60				
66		34	1535				
02		MHZ			26SW/1544	ZHU	

“...REPORT LEAVING SIX THOUSAND...”

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Phraseology Example

RR REPORT REACHING							
N45T	GLH 1600	12	110	IGB 1644	KTXK V278 IGB KUBS/1705		
BE65/A T175		16	↓60 RR				
66		12					
02		SQS					

“...REPORT REACHING SIX THOUSAND...”

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⦿ Used in space 20 with altitude reports

- RL – Report Leaving
- RR – Report Reaching

Continued on the next page

CLEARANCE AND MISCELLANEOUS ABBREVIATIONS

(Continued)

Request for Altitude/Fix Reports (Cont'd)

JO 7110.65,
par. 2-3-10; JO
7340.2

RP		REPORT PASSING	
RX		REPORT CROSSING	
N234	MLU 1600 1606	13	110✓
C310/A T180	16		MHZ
66	13	1613	KMLU V18 MHZ KJAN/1645
01	STUEE		RP DINKY/ <i>Time</i> RX SQS194R/ <i>Time</i>

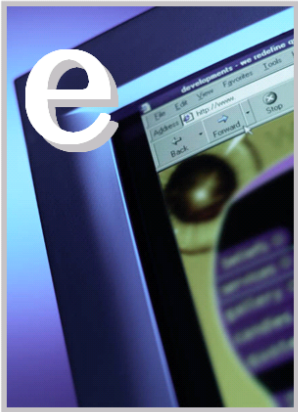
20

"TWIN CESSNA TWO THREE FOUR, REPORT PASSING DINKY INTERSECTION, REPORT CROSSING SIDON ONE NINER FOUR RADIAL."

- ⊙ Used in space 26 with fix, radial, or DME reports
 - RP – Report Passing
 - RX – Report Crossing
- ⊙ When the report is received from the pilot, the current time is recorded after the slant (/) in four digits.

ACTIVITY 1: IDENTIFYING ABBREVIATIONS

Activity 1



IDENTIFYING ABBREVIATIONS ACTIVITY

Purpose: to practice identifying and using the clearance and miscellaneous abbreviations used in stripmarking

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Description

This eLearning activity will provide you with a chance to practice identifying the clearance and miscellaneous abbreviations used in stripmarking. You will answer a total of 21 questions by selecting applicable abbreviations from the image displayed.

Directions

Access the IET eLearning menu. Select **Lesson 6 – Recording Clearances and Control Information**. Click on the title to launch the **Identifying Abbreviations** activity.

Turn on the Cap Lock feature on your keyboard.

Time Allotted

10 minutes

CONTROL INFORMATION SYMBOLS

Control Information Symbols

JO 7110.65,
par. 2-3-10

⊙ Control information symbols include:

T → <i>sw</i>	Depart (Direction, if Specified)
/	Until
⇒	Join or Intercept Airway/Jet Route/Track or Course
—	Restriction Bar
↑ ↓	Climb/Descend and Maintain Arrows
X	Cross
↑ or ↓	At or Above/Below
— (Dash)	From-To
V < (time)	Clearance Void if Aircraft not off Ground by (Time)
()	Alternate Instructions
C	Communications Transfer
> <	Before and After
→	Cruise
↗ ↘	Direction of Flight Indicator
(alt)	Aircraft Reported at Other than Assigned Altitude
<u>alt</u> (underline in red)	Inappropriate Altitude for Direction of Flight (IAFDOF)
✓	Aircraft Reported at Assigned Altitude
E (red)	Emergency
W (red)	Warning

Continued on next page

CONTROL INFORMATION SYMBOLS *(Continued)*

Control Information Symbols (Cont'd)

JO 7110.65,
par. 2-3-10

 (red)	Information Revised/Information Forwarded
	Pilot Canceled Flight Plan
(alt) B (alt)	Block Altitude Assignment
R (alt)	Requested Altitude
R	Radar Contact
	Radar Service Terminated
	Radar Contact Lost
	Radar Handoff
RV	Radar Vector
	Pilot Resumed Own Navigation
P	Point Out Initiated
	Enter Control Area

Continued on next page

CONTROL INFORMATION SYMBOLS *(Continued)*

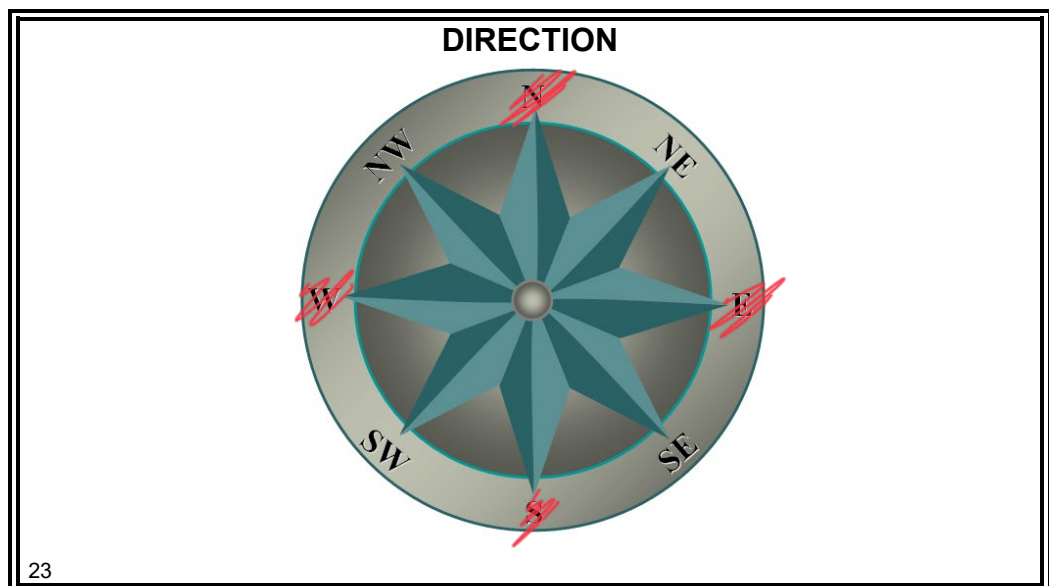
**Depart
(Direction, if
Specified)**
JO 7110.65,
pars. 2-3-10, 4-3-2

T → DEPART (DIRECTION, IF SPECIFIED)					
N456 C182/A T160 66 01	T→SW TR 240/ ⇒ V278 1355/ KGWO P1400	↑	GLH 60	KGWO SQS V278 KGLH/0014	D-A

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- ⊙ Used **only** at airports within Class D or E surface areas
- ⊙ Used in space 15 on departure strip **only**

Direction
JO 7110.65,
pars. 2-3-10, 4-3-2



- ⊙ Direction of departure is specified:
 - To provide separation
 - Using eight compass points

CONTROL INFORMATION SYMBOLS *(Continued)*

Until

JO 7110.65,
par. 2-3-10

/ UNTIL						
N34407		T→SW TL150 / →V278	↑90	IGB	KGWO SQS V278 IGB KUBS/0050	D-A
PA31/A T185						
66		1357/				
01		KGWO P1400	↑90	90		
24						

⊙ Used:

- In any space
- Usually with (time)/(fix)
- As separator between route of flight and ETA/ETE for general aviation aircraft

Join or Intercept Airway/Jet Route/Track or Course

JO 7110.65,
par. 2-3-10

⇒ JOIN OR INTERCEPT AIRWAY/JET ROUTE/TRACK OR COURSE						
N511WC		T→SW TR340 ⇒ V535	↑70	HLI	KGWO SQS V535 HLI KMEM/0020	D-A
PA31/A T180						
66		1229/				
01		KGWO P1230	↑70	70		
25						

⊙ Used in spaces 15 or 25

CONTROL INFORMATION SYMBOLS *(Continued)*

Restriction Bar

JO 7110.65,
par. 2-3-10

RESTRICTION BAR						
N15T		T→SW - SQS	↑80	MHZ	KGWO SQS V557 MHZ KJAN/0023	D-A
C310/A			X17SW SQS			
T180			↓50			
66		1425/	X17NW MHZ			
01		KGWO P1430	↑60	80		

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- ⊙ Used in space 20 to separate altitude assignments from altitude restrictions
 - Restrictions on how to get to assigned altitude

Climb/ Descend and Maintain Arrows

JO 7110.65,
par. 2-3-10

CLIMB/DESCEND AND MAINTAIN ARROWS						
UAL42	IGB	24	400	120	GLH	KGTR IGB V278 KGLH
B721/A	0111	01	↓	0129		
T420						
66						
02		SQS				

DAL61	HEZ	10	130	110	KJAN	KBTR V245 MHZ KJAN
B721/A	0100	01	↓			
T420						
66						
01		MHZ				

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- ⊙ Used in space 20

Continued on next page

CONTROL INFORMATION SYMBOLS *(Continued)*

Cross

JO 7110.65,
par. 2-3-10

X CROSS						
N15T C310/A T180 66 01		↑	↑120	SQS	KJAN MHZ V9 UJM KHEE/0058	D-A
			X06SE SQS@			
			120			
			1425/			
			KJAN P1430	120		

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- ⊙ Used in space 20 to cross a fix/radial/airway
 - Followed by a restriction

At or

Above/Below

JO 7110.65,
par. 2-3-10

↑ or ↓ AT OR ABOVE/BELOW						
N15T C310/A T180 66 01		↑	↑120	SQS	KJAN MHZ V9 UJM KHEE/0058	D-A
			X17NW MHZ ↑			
			X06SE SQS			
			10			
			1425/			
			KJAN P1430	120		

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- ⊙ Used in space 20 followed by an altitude

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CONTROL INFORMATION SYMBOLS *(Continued)*

From-To

JO 7110.65,
par. 2-3-9, 2-3-10

FROM - TO					
N124	T→SW ← QS	↑70	HLI	KGWO SQS V535 HLI KMEM/0040	D-A
P28A/A T150	1355/				
66 01	KGWO P1400		70		

“...VIA DEPART SOUTHWEST DIRECT SIDON”

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- ⊙ When used in space 15 and 25, it is spoken as “direct.”
- ⊙ When used in space 28, the dash follows an appropriate control symbol and precedes pertinent control information.

Clearance Void if Not Off by (Time)

JO 7110.65,
par. 2-3-10

V < (TIME) CLEARANCE VOID IF NOT OFF BY (TIME)					
N124	T→NE TR030 / = V417 V < 1515 (20)	↑70 X20 SW MHZ ↑ 60	MHZ	KVKS MHZ V18 KMEI/0040 V417	D-A
P28A/A T150	1458/				
66 01	KVKS P1500		70		

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- ⊙ Used:
 - In space 15 followed by a time
 - To avoid delay for other traffic at airports:
 - Where communications with aircraft are difficult until airborne
 - **Not** served by control towers
- Provide alternate instructions requiring pilots to advise ATC of intentions

Continued on next page

CONTROL INFORMATION SYMBOLS *(Continued)*

Alternate Instructions

JO 7110.65,
par. 2-3-10



Phraseology Example

() ALTERNATE INSTRUCTIONS						
N124		T→NE TR030 / ⇒ V417 V < 1515 (20)	↑70 X20 SW MHZ ⚡ 60	MHZ	KVKS MHZ V18 KMEI/0040 V417	D-A
P28A/A T150 66 01		1458/				
		KVKS P1500		70		

“...CLEARANCE VOID IF NOT OFF BY ONE FIVE ONE FIVE. IF NOT OFF BY ONE FIVE ONE FIVE, ADVISE AERO CENTER NOT LATER THAN ONE FIVE TWO ZERO OF INTENTIONS.”

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⊙ Used in space 15 here, can be used in other spaces.

Continued on next page

CONTROL INFORMATION SYMBOLS *(Continued)*

Communications Transfer

JO 7110.65,
par. 2-3-10

C COMMUNICATION TRANSFER						
N456	GLH	46	90✓	HLI	KGLH V278 SQS V11 HLI	
PA31/A	1634	16		1706	M41/1736	
T180		45				
66		1645				
02		SQS				
					25NE	

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- ⊙ Used in space 26
 - Include time, fix, or altitude unless compliance is expected upon receipt
 - Insert frequency when other than standard
 - Frequency is assigned when clearance is issued through:
 - FDU
 - Tower, unless covered in LOA
- ⊙ Approach Controls will assign frequency as part of communications transfer
 - Provided it is covered in LOA

CONTROL INFORMATION SYMBOLS *(Continued)*

Before and After

JO 7110.65,
par. 2-3-10



Phraseology Examples

> < **BEFORE AND AFTER**

N63T BE35/A T180 66 02	MHZ 1604	20 16 19 HATER	↓ 80↓70 70>1625	MLU	KJAN MHZ V427 MLU KMLU/1635
---	-------------	--	---	-----	--------------------------------

"...DESCEND SO AS TO REACH SEVEN THOUSAND
BEFORE ONE SIX TWO FIVE..."

N71C C182/U T140 66 01	T → SW TR 240 / ⇒ V278 RLS 1MIN < N21 1644/ KGWO P1645	↑ ↑60	GLH	KGWO SQS V278 KGLH/0014	D-A
---	--	---	-----	----------------------------	--

"...RELEASE ONE MINUTE AFTER NOVEMBER TWO ONE..."

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⊙ Used in any space

Continued on next page

CONTROL INFORMATION SYMBOLS *(Continued)*

Cruise

JO 7110.65,
par. 2-3-10



Phraseology Example

CRUISE

N234 C310/A T180 66 01	MLU 1638	54 ↓		60✓	VKS 1658	KMLU V417 DORTS VKS KVKS/1658	APCH 1654
		16		→			
		54	DORTS				

“TWIN CESSNA TWO THREE FOUR CLEARED TO
VICKSBURG AIRPORT, CRUISE SIX THOUSAND.”

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⦿ Used in space 20

Direction of Flight Indicator

JO 7110.65,
par. 2-3-10



DIRECTION OF FLIGHT INDICATOR

AAL21 B721/A T420 66 01	HEZ 1056	06 11		170✓	ZAMMA	KMSY V245 IGB KGTR	
		MHZ					
		→					

DAL357 B732/A T420 66 02	IGB 1656	08 17		170✓	MHZ	KGTR IGB V278 SQS V555 MHZ KJAN	
		07					
		SQS					

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⦿ Used in space 23

Continued on next page

CONTROL INFORMATION SYMBOLS *(Continued)*

**Aircraft
Reported at
Other Than
Assigned
Altitude**
JO 7110.65,
par. 2-3-10

AIRCRAFT REPORTED AT OTHER THAN ASSIGNED ALTITUDE							
N234	MLU	30	110	MHZ	KMLU V427 MHZ KJAN/1643		
C310/A T180	1615	16	120✓				
66		30					
01		HATER					

The aircraft was assigned 110, but reported on frequency level at 120.

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NOTE: In some cases, an aircraft may have been assigned an altitude by another controller without proper coordination.

- ⦿ Used in space 20
 - Write reported altitude and circle it in black

**Inappropriate
Altitude for
Direction of
Flight
(IAFDOF)**
JO 7110.65,
par. 2-3-10

<u>alt</u> INAPPROPRIATE ALTITUDE FOR DIRECTION OF FLIGHT							
N357	HEZ	39	80✓	KJAN	KCRP HEZ V245 MHZ KJAN/1810		
C182/A T140	1701	17	↓				
66							
01		MHZ					

38

- ⦿ Used in spaces 20 or 24
- ⦿ Underlined in red

CONTROL INFORMATION SYMBOLS *(Continued)*

Aircraft Reported at Assigned Altitude

JO 7110.65,
par. 2-3-10

AIRCRAFT REPORTED AT ASSIGNED ALTITUDE

DAL21	MLU	18	14	196 ✓ ✗170 ✓	MHZ	KTUL MLU V18 MHZ V245 IGB KUBS	
B721/A T420	1415		19 1418				
66			STUEE				
01							

39

- ⦿ Used in space 20

Emergency

JO 7110.65,
par. 2-3-10

EMERGENCY

UAL56	HEZ	49	16	150	KJAN	KHOU./HEZ V245 MHZ KJAN	
A319/A T420	1627						
66			49			E #2 ENGINE OUT	
01			MHZ				

40

- ⦿ Used in space 26 at Aero Center
- ⦿ Written in red
 - Information describing the emergency is written in black

Continued on next page

CONTROL INFORMATION SYMBOLS *(Continued)*

Warning
JO 7110.65,
par. 2-3-10

 WARNING EN ROUTE STRIP						
N356 C310/A T180 66 02	HEZ 1618	30 16	90 	ZAMMA	KHOU./HEZ V245 IGB KUBS/1706	
		MHZ				
PROPOSAL STRIP						
N711JI PAY1/A T230 66 01		↑		HLI	KGWO SQS V11 HLI KSTL/0112	
		KGWO P2215		90 		

41

- ⊙ Used in space 20 to alert controller that **action must** be taken
- ⊙ Usually **adjacent to altitude**
- ⊙ Written in **red**

Continued on next page

CONTROL INFORMATION SYMBOLS *(Continued)*

Information or Revised Information Forwarded

JO 7110.65,
par. 2-3-10


INFORMATION OR REVISED INFORMATION FORWARDED

N234	IGB 1144	12	13	100	GLH	KUBS IGB V278 KGLH /1228
C310/A T180		18	120			
66		18				
02		SQS				

Sector 66 forwards to Sector 67.

42

- ⊙ Used in any space
- ⊙ Circle control information in red when forwarded

Pilot Canceled Flight Plan

JO 7110.65,
par. 2-3-10


PILOT CANCELED FLIGHT PLAN

N234	MHZ 1618	30	80✓	VKS 1633	KMEI V18 MHZ V417 DORTS VKS KVKS/1633
C310/A T180		16 ↓ 30 1628			
66		1628			
03		DORTS			

43

- ⊙ Used in space 18 with a four-digit time

Continued on next page

CONTROL INFORMATION SYMBOLS *(Continued)*

Block Altitude Assignment

JO 7110.65,
par. 2-3-10

(alt) B (alt) BLOCK ALTITUDE ASSIGNMENT							
A69843	TXK	20	190B230	IGB	KTXX J52 IGB KCBM	2321	
H/C141/A T450 G450	1457	15					
66							
105	01	SQS					

44

- ⊙ Used in space 20
- ⊙ Altitudes are inclusive
 - First altitude **must** be lower than the second

Radar Contact

JO 7110.65,
par. 2-3-10

R RADAR CONTACT							
UAL21	DORTS	26	170	KJAN	KMLU V417 MHZ KJAN	2776	
B732/A T420 G420	1419	14					
66							
234	02	MHZ					

45

- ⊙ Normally used in spaces 22, 23, or 24
 - Space may be locally adapted for each facility

Continued on next page

CONTROL INFORMATION SYMBOLS *(Continued)*

Radar Service Terminated

JO 7110.65,
par. 2-3-10

 RADAR SERVICE TERMINATED		
VKS	KJAN MHZ V417 DORTS VKS KVKS/1650	7121
		

46

Radar Contact Lost

JO 7110.65,
par. 2-3-10

 RADAR CONTACT LOST		
GLH	KJAN MHZ V74 KGLH/1455	2273
		

47

Continued on next page

CONTROL INFORMATION SYMBOLS *(Continued)*

Radar Handoff
JO 7110.65,
par. 2-3-10

Ⓡ RADAR HANDOFF		
HLI	KJAN MHZ V11 HLI M41	3332

48

- ⦿ Circle "R" in black when handoff is completed.

Radar Vector
JO 7110.65,
par. 2-3-10

RV RADAR VECTOR		
MCB	KMEM V9 KMCB	7214

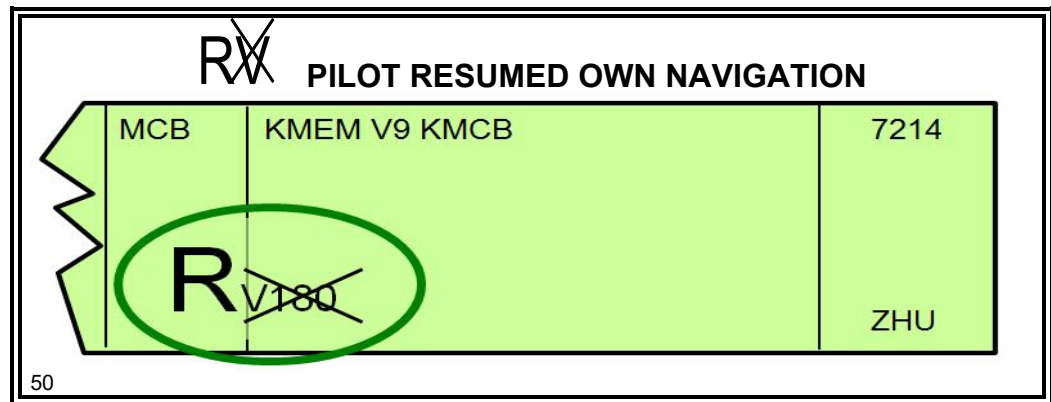
49

- ⦿ Add "V" after "R" when assigning a vector to an aircraft.
 - Assigned heading follows the "V"

Continued on next page

CONTROL INFORMATION SYMBOLS *(Continued)*

**Pilot Resumed
Own
Navigation**
JO 7110.65,
par. 2-3-10

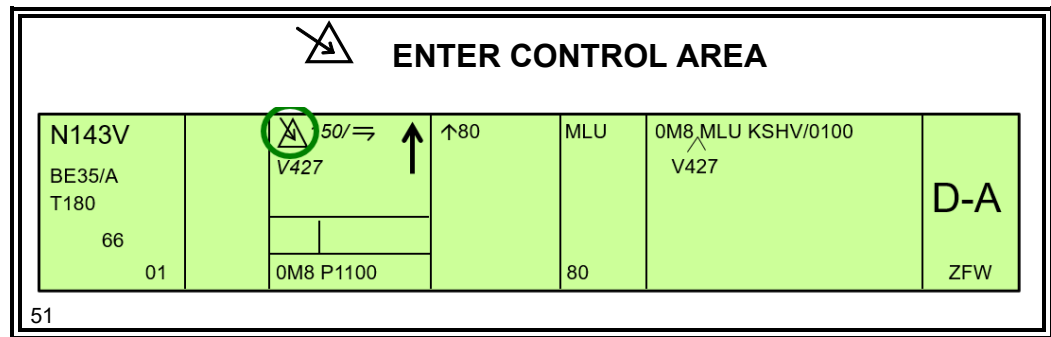


- ⦿ Draw an X through "V" and heading when the aircraft is taken off the vector and returned to its own navigation.

CONTROL INFORMATION SYMBOLS *(Continued)*

Enter Control Area

JO 7110.65,
par. 2-3-10



⦿ Used:

- In space 15
- For aircraft requesting clearance in other than Class A, B, C, D, and E areas

Continued on next page

CONTROL INFORMATION SYMBOLS *(Continued)*

Knowledge Check

❖ **QUESTION:** How is the word “depart” recorded in space 15?

- A. T
- B. D
- Ⓒ. T→

52

❖ **QUESTION:** What does the abbreviation/symbol T→N TL in space 15 mean?

- A. “Take off north, left turn.”
- B. “Depart now, turn left.”
- Ⓒ. “Depart north, turn left.”

53

Continued on next page

CONTROL INFORMATION SYMBOLS *(Continued)*

Knowledge
Check
(Cont'd)

QUESTION: An aircraft has been cleared from 140 to 80 and requested to report leaving 140 and reaching 80. How and where should these reports be recorded?

54

140 ✓ ↓ 80
RL 140
RL 80

QUESTION: When an aircraft reports at an altitude other than its assigned altitude, the altitude is _____.

- A. Circled in red
- B. Circled in black**
- C. Underlined in black

55

APPROACH ABBREVIATIONS

Approach Abbreviations

JO 7110.65,
par. 2-3-10;
JO 7340.2



Phraseology Example

APPROACH ABBREVIATION						
N234	MLU 1618	30 ↓	110✓ ↘	VKS 1635	KMLU V417 DORTS VKS KVKS /1635	APCH 1624
C310/A T180		16				
66		29				
01		DORTS				

“TWIN CESSNA TWO THREE FOUR, CLEARED APPROACH VICKSBURG AIRPORT.”

56

⦿ APCH – Approach

- Used in space 28
 - Circle in red when coordinated
 - Write in black when aircraft is cleared for approach
 - Including four-digit time when aircraft was cleared for approach
- Implies pilot may execute approach of his/her choice

Specific Approach Procedure Abbreviations

JO 7110.65,
par. 2-3-10

- ⦿ The following specific approach procedure abbreviations are entered in space 28:
 - VR – VOR approach
 - ILS – ILS approach
 - NDB – Nondirectional Radio Beacon approach
 - VA – Visual approach
 - GPS – GPS approach

EXERCISE 1: RECORDING CLEARANCES AND CONTROL INFORMATION

Exercise 1



RECORDING CLEARANCES AND CONTROL INFORMATION EXERCISE



Purpose: to practice recording clearances and control information on flight strips

Directions: record the clearances and control information on the flight progress strips using the clearance/request provided

57

Directions

For items 1-13, use approved stripmarking and record the clearances and control information on the flight progress strips using the clearance/request preceding each strip.

Continued on next page

EXERCISE 1: RECORDING CLEARANCES AND CONTROL INFORMATION *(Continued)*

Questions

1. "Queen Air Five Six Seven, cleared to St. Louis Airport via depart southwest, turn right fly heading two niner zero until joining Victor Niner, Victor Niner. Cross niner miles northwest Sidon VORTAC established on Victor Niner at or below six thousand, climb and maintain one zero thousand."

EXERCISE 1 – QUESTION 1					
N567		↑	UJM	KGWO SQS V9 KSTL/0215	
BE80/A					
T180					
66					
01		KGWO P1300	100		

58

2. "Cessna Two Three Four, cleared to Leakesville Airport via direct Sidon, Victor Eleven. Cross one seven miles southeast Sidon VORTAC at or below six thousand, cross one seven miles northwest Magnolia VORTAC at or above eight thousand, climb and maintain one one thousand."

EXERCISE 1 – QUESTION 2					
N234		↑	↑ 110	MHZ	KGWO SQS V11 KGCV/0055
C310/A			X 17 SE SQS		
T200			↓ 60		
66			X 17 NW MHZ		
01		KGWO P1630	↑ 80	110	D - A

59

Continued on next page

EXERCISE 1: RECORDING CLEARANCES AND CONTROL INFORMATION *(Continued)*

Questions (Cont'd)

3. "Queen Air Two Five Six Quebec, cleared from Vicksburg Airport to Meridian Airport, via depart northeast turn right, fly heading zero three zero until joining Victor Four Seventeen, Victor Four Seventeen Magnolia then as filed, Cross two zero miles southwest Magnolia VORTAC established on Victor Four Seventeen at or above six thousand, climb and maintain niner thousand. Clearance void if **not** off by one four one zero. If **not** off by one four one zero advise Aero Center **not** later than one four one five of intentions. Verify this clearance will allow compliance with local traffic pattern and terrain or obstruction avoidance. Advise, Queen Air Two Five Six Quebec Released for Departure, Contact AERO Center One Two Five Point Zero."

EXERCISE 1 – QUESTION 3					
N256Q		T → NE TR 030 / 5 V417	↑ 90 X 20 SW MHZ A 60	MHZ	KVKS MHZ V18 KMEI/0125
BE65/A T175		V < 1410 (15)			
66	EDL	1400/			
01	1405	KVKS P1400	↑ 90	90	D-A

60

Continued on next page

EXERCISE 1: RECORDING CLEARANCES AND CONTROL INFORMATION *(Continued)*

Questions (Cont'd)

4. "Cessna Five Six Three Juliett, cleared from Vicksburg Airport to East Texas Regional Airport via depart northeast, turn left, fly heading three three zero until joining Victor Four Seventeen, Victor Four Seventeen Monroe, then as filed, Cross three one miles southeast Monroe VORTAC established on Victor Four Seventeen at or above seven thousand. Climb and maintain one two thousand. Hold for release."

EXERCISE 1 – QUESTION 4						
N563J		T → NE TL 330 / 3 V417	↑ 120 X 31 SE MLU ↑ 70	MLU	KVKS MLU V18 KGGG/0141	D-A
C310/A		HFR				
T180		 /				
66	EDL 1635	KVKS P1630		120		ZFW
01						
61						

Continued on next page

EXERCISE 1: RECORDING CLEARANCES AND CONTROL INFORMATION *(Continued)*

Questions (Cont'd)

5. "Delta Twenty-One, cleared to Magnolia VORTAC, descend and maintain six thousand, hold northwest as published, **no** delay expected. Contact Jackson Approach one one niner point two at one two three zero."

EXERCISE 1 – QUESTION 5						
DAL21 B721/A T450 66 01	HEZ 1220	32 ↓ 12 31 MHZ	170✓ ↓ 60	KJAN	KHOU HEZ V245 MHZ KJAN <i>C1230 or C245V MHZ miles</i>	H-mw
62						

6. "Aero Center, Cessna Two Niner One One Echo estimating Sidon VORTAC two one one niner at one zero thousand."

EXERCISE 1 – QUESTION 6						
N2911E C182/A T170 66 02	IGB 2050	20 21 SQS	120	GLH	KUBS IGB V278 KGLH/2133	
63						

Continued on next page

EXERCISE 1: RECORDING CLEARANCES AND CONTROL INFORMATION *(Continued)*

Questions (Cont'd)

7. "United Five Zero One, cleared to Meridian Airport via last routing cleared."

EXERCISE 1 – QUESTION 7						
UAL501	HEZ	32	190✓	MEI	KHOU HEZ V245 MHZ V18 KMEI	H NW
DC91/A T420	1221	12				
66		31				
01		MHZ				

64

8. "Air Force Six Seven Niner Three Four, maintain one two thousand until one four miles northeast Sidon VORTAC, cross Sidon VORTAC at or above one three thousand, climb and maintain block flight level one eight zero through flight level two zero zero."

EXERCISE 1 – QUESTION 8						
A67934	IGB	20	120✓	GLH	KCBM IGB V278 GLH KLRF	
3/C130/A T300	2103	21				
66		20				
02		SQS				

65

Continued on next page

EXERCISE 1: RECORDING CLEARANCES AND CONTROL INFORMATION *(Continued)*

Questions (Cont'd)

9. "Cessna Six Two One One Four, cleared to McComb Airport via direct Sidon, Victor Niner. Climb and maintain one zero thousand."

EXERCISE 1 – QUESTION 9							
N62114			↑ 100	MHZ	SQS V9 MCB KMCB/0045		D-A
C182/A							
T150							
66							
692	01		SQS P1400	100	PICK UP OVER SQS		

66

10. "Delta Thirty-Seven, cleared to Memphis Airport via direct SQS as filed. Climb and maintain one zero thousand, expect one seven thousand one zero minutes after departure."

EXERCISE 1 – QUESTION 10							
DAL37			↑ 100	HLI	KGWO SQS V535 HLI KMEM		D-A
MD88/A							
T450							
66							
142	01		KGWO P1630	(170	10 < 0)		

67

Continued on next page

EXERCISE 1: RECORDING CLEARANCES AND CONTROL INFORMATION *(Continued)*

Questions (Cont'd)

11. "November Five Six Six maintain seven thousand until one three miles northwest of Sidon VORTAC, cleared VOR runway five approach circle to runway two three."
TIME: 0124

EXERCISE 1 – QUESTION 11						
N566 BE80/A T180 66 02	GLH 0118	30	70✓ 70/15 NW	KGWO 0137	KGLH V278 SQS KGWO/0135	VR 0124
		01				
		29				
		SQS			67 70	

68

NOTE: Arrival information **must** be forwarded to GWO Tower before approach clearance is issued. This is indicated on the strip by circling the GWO estimate in red and by writing and circling VR in red in space 28.

Continued on next page

EXERCISE 1: RECORDING CLEARANCES AND CONTROL INFORMATION *(Continued)*

Questions (Cont'd)

12. "November Five Six Juliett Sierra, cleared to Magnolia VORTAC, maintain one one thousand until two zero miles northeast Natchez VOR/DME, descend and maintain six thousand, hold northwest as published, **no** delay expected.

EXERCISE 1 – QUESTION 12						
N56JS	HEZ 1007	41 ↓	110 ✓ ↓ 60 110 / 20 N E 462	KJAN	KAEX V245 MHZ KJAN/1041	H- ^{nw}
C310/A T200		10				
66		40				
02		MHZ				

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Continued on next page

EXERCISE 1: RECORDING CLEARANCES AND CONTROL INFORMATION *(Continued)*

Questions (Cont'd)

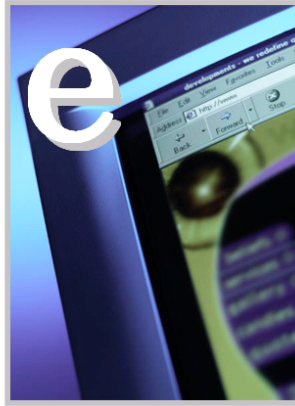
13. "Cessna Niner Six Seven Bravo cleared from the Vicksburg Airport to the Alexandria Airport via the Natchez Zero Two Six Radial, Natchez, then as filed. Climb and maintain eight thousand, report leaving six thousand. Clearance void if not off by one four one zero. If not off by one four one zero, advise Aero Center not later than one four one five of intentions. Verify this clearance will allow compliance with local traffic pattern and terrain or obstruction avoidance. Advise, Cessna Niner Six Seven Bravo Released for Departure, Contact AERO Center One Two Five Point Zero."

EXERCISE 1 – QUESTION 13						
N967B			↑ 80	HEZ	KVKS HEZ KAEX/0040	
C208/A		↑			<u>HEZ 01R</u>	D-A
T160		V < 1410 (15)				
66	EDL	1400				
01	1405	KVKS P1400	RL 60	↑ 80	80	ZHU
70						

ACTIVITY 2: IDENTIFYING CONTROL INFORMATION SYMBOLS

Activity 2

IDENTIFYING CONTROL INFORMATION SYMBOLS ACTIVITY



Purpose: to practice using control information symbology used in stripmarking

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Description

This exercise provides you with opportunities to practice identifying the control information symbols used in stripmarking. The first activity consists of a set of electronic flash cards that allows you to test yourself on how well you know the symbols. You will have one flash card per symbol and will click the card to view the description of the symbol.

Directions

Access the IET eLearning menu. Select **Lesson 6 – Recording Clearances and Control Information**. Click on the title to launch the **Identifying Control Information Symbols** activity.

Time Allotted

10 minutes

RECORDING REPORTS/TIMES ON FLIGHT PROGRESS STRIPS

Position/DME Reports

JO 7110.65,
par. 2-3-2

POSITION/DME REPORTS							
N291EE	MHZ 1526	56 ↓	70✓	KGWO 1603	KJAN MHZ V9 SQS KGWO/1600	VR	H SW -256 LT 1605
C182/A T170		15					
66		55					
02		SQS			20 SE/1550		

72

☉ Record reports in space 26 followed by:

- Slant (/)
- Time reported by pilot

Times

JO 7110.65,
par. 2-3-2

PREVIOUS FIX							
	11						
	12						
	13						
	14						
	14a						

73

☉ Spaces 11 – 14

- Space 11 - Previous fix
- Space 12 - Previous fix estimated time
- Space 13 - Revised previous fix estimated time
- Space 14 - Previous fix actual time
 - Actual departure time on first fix posting after departure
- Space 14a - Plus time (in minutes) from previous fix to posted fix

Continued on next page

Times
(Cont'd)
JO 7110.65,
par. 2-3-2

- ⊙ Spaces 15, 17, 18, and 19
 - Space 15 - Center-estimated time over posted fix / *Estimated Stop* *Signature Instructions*
 - Space 17 - Pilot-estimated time over posted fix
 - Written in two digits (minutes)
 - Space 18
 - Actual time (pilot-reported) over posted fix
 - Time entered/leaving holding fix
 - Arrival time (at nonapproach control airport) if posted fix in space 19 is the airport.
 - Cancellation of IFR
 - Departure time
 - Actual (written in black), or
 - Assumed (written in red)
- ⊙ Space 19 - Posted fix

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RECORDING REPORTS/TIMES ON FLIGHT PROGRESS STRIPS *(Continued)*

Times
(Cont'd)
JO 7110.65,
par. 2-3-2

NEXT POSTED FIX						
			21			
			22			

- ① Spaces 21 and 22
 - Space 21 - Next posted fix, write airport ID, if not already in space 19 or 21.
 - Space 22 - Pilot's estimate at next fix, or arrival time (at non approach control airport) if airport in space 21.
 - Pilot will give estimate of this fix when they report progressing the posted fix

EXERCISE 2: STRIPMARKING

Exercise 2



STRIPMARKING EXERCISE



Purpose: to practice marking flight progress strips

Directions: complete the strips based on information provided by instructor

76

Directions

Your instructor will read 14 clearances aloud. You should record each clearance on a different flight progress strip provided by your instructor.

Continued on next page

EXERCISE 2: STRIPMARKING *(Continued)*

Flight Strips

1. JAN Approach requests clearance, Delta One-Fifty, to MSY Airport.
(Coordination has been completed.)

EXERCISE 2 – QUESTION 1						
DAL150 MD82/A T460 66 01			↑		MCB 140	KJAN MHZ V9 MCB KMSY/0025 ZHU
77						

Continued on next page

EXERCISE 2: STRIPMARKING *(Continued)*

Flight Strips (Cont'd)

2. Greenwood Tower requests clearance Blue Streak Two Twenty-One, to Memphis Airport. (Coordination has been completed.)

EXERCISE 2 – QUESTION 2									
JIA221			↑		HLI	KGWO SQS V535 HLI KMEM			
E120/A									
T300									
66									
01			KGWO P1200		150				

78

NOTE: Assumed departure time is written in red in space 18.

Continued on next page

EXERCISE 2: STRIPMARKING *(Continued)*

Flight Strips (Cont'd)

3. Aero Center Flight Data requests clearance, Twin Cessna Four Six Three, from Vicksburg Airport to Meridian Airport. (Coordination has been completed.)

EXERCISE 2 – QUESTION 3						
N463		↑		MHZ	KVKS MHZ V18 KMEI/0038	
C310/A						
T180						
66						
01		KVKS P1600		90		

Continued on next page

EXERCISE 2: STRIPMARKING *(Continued)*

Flight Strips (Cont'd)

4. N156 is estimating Sidon VORTAC two three five zero, level at one one thousand.

EXERCISE 2 – QUESTION 4						
N156	MHZ 2330	49 23	↓	110	KGWO 2352	KMSY./MCB V9 SQS KGWO/2352
PA27/A T180						
66						
02		SQS				
80						

- 5.

EXERCISE 2 – QUESTION 5						
N234	IGB 1601	30 16		100	GLH	KBNA V278 GLH KTXK/1736
C310/A T180						
66						
04		SQS				
81						

N234	IGB 1601	30 16		100✓	GLH	KBNA V278 GLH KTXK/1736
C310/A T180						
66						
04		29 SQS				

Continued on next page

EXERCISE 2: STRIPMARKING *(Continued)*

Flight Strips (Cont'd)

6. N147 is estimating Sidon VORTAC one six three one, level at nine thousand.

EXERCISE 2 – QUESTION 6						
N147	MHZ	30	90	KGWO	KMSY V9 SQS KGWO/1634	
BE35/A	1610	16				
T180						
66						
02		SQS				

82

7. Indicate that AAL16 estimated Magnolia VORTAC one six two seven, at one seven thousand and has declared an emergency due to a fuel leak.

EXERCISE 2 – QUESTION 7						
AAL16	STUEE	27	170	KJAN	KDFW./MLU V18 MHZ	
MD88/A	1618	16			KJAN	
T450						
66						
02		MHZ				

83

Continued on next page

EXERCISE 2: STRIPMARKING *(Continued)*

Flight Strips
(Cont'd)

8.

EXERCISE 2 – QUESTION 8						
AAL431	SQS 1519	26	160✓	HEZ	KMEM UJM V9 SQS V557 MHZ V245 HEZ KDFW	ZHU
MD82/A T450		15				
66						
04		MHZ				

84

9.

EXERCISE 2 – QUESTION 9						
DAL412	MCB 2121	30	150	SQS	KMSY V9 SQS V535 HLI KMEM	
B753/A T480		21				
66						
01		MHZ				

85

Continued on next page

EXERCISE 2: STRIPMARKING *(Continued)*

Flight Strips
(Cont'd)

10.

EXERCISE 2 – QUESTION 10						
N16S	ZAMMA	30	120	STUEE	KUBS IGB V245 MHZ V18	
BE35/A	1615	16			MLU KMLU/1703	
T180						
66						
02		MHZ				
86						

Continued on next page

EXERCISE 2: STRIPMARKING *(Continued)*

Flight Strips (Cont'd)

11. N521 is level at one one thousand.

EXERCISE 2 – QUESTION 11						
N521	MHZ 2108	27 ↓	110✓	KGWO 2134	KMSY HEZ V245 MHZ V9 SQS KGWO/2135	
BE35/A T180		21				
66		27				
02		SQS				
					67↕70	

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Continued on next page

EXERCISE 2: STRIPMARKING *(Continued)*

Flight Strips
(Cont'd)

12.

EXERCISE 2 – QUESTION 12						
N45T	HLI 1550	12	100	MHZ	M41 HLI V535 SQS V11	
BE65/U T175		16			MHZ KJAN/1632	
66						
02		SQS				
88						

Continued on next page

EXERCISE 2: STRIPMARKING *(Continued)*

Flight Strips (Cont'd)

13. Greenwood Tower requests clearance Lear Two Two Lima Juliett, to Atlanta Airport. (Coordination has been completed.)

EXERCISE 2 – QUESTION 13							
N22LJ			↑		IGB	KGWO SQS J52 IGB RMG KATL/0130	
LJ35/A							
T450							
66							
01			KGWO P1800		230		
89							

NOTE: Assumed departure time is written in red in space 18.

Continued on next page

EXERCISE 2: STRIPMARKING (Continued)

Flight Strips
(Cont'd)

14.

EXERCISE 2 – QUESTION 14						
N125X C310/A T180 66 01	MCB 1245	10		130✓	MEI	KMCB V557 MHZ V18 KMEI/1357
		13				
		10				
		MHZ				

90

IN CONCLUSION

Lesson Review

LESSON REVIEW

The following topics were covered in this lesson:

- Control symbology
- Clearance and miscellaneous abbreviations
- Control information symbols
- Approach abbreviations
- Recording reports/times on flight progress strips



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End-of-Lesson Test

END-OF-LESSON TEST

**Recording
Clearances and
Control
Information**



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APPENDIX A: ABBREVIATIONS TABLE

Abbreviations

JO 7110.65,
par. 2-3-10, table
2-3-11

Abbreviation	Meaning
A	Cleared to airport (point of intended landing)
B	Center clearance delivered
C	ATC clears (when clearance relayed through non-ATC facility)
CAF	Cleared as filed
D	Cleared to depart from the fix
F	Cleared to the fix
H	Cleared to hold and instructions issued
L	Cleared to land
N	Clearance not delivered
O	Cleared to the outer marker
PD	Cleared to climb/descend at pilot's discretion
Q	Cleared to fly specified sectors of a NAVAID defined in terms of courses, bearings, radials or quadrants within a designated radius.
T	Cleared through (for landing and takeoff through intermediate point)
V	Cleared over the fix
X	Cleared to cross (airway, route, radial) at (point)
Z	Tower jurisdiction

Continued on next page

APPENDIX A: ABBREVIATIONS TABLE *(Continued)*

Miscellaneous Abbreviations

JO 7110.65,
par. 2-3-10, table
2-3-12

Abbreviation	Meaning
BC	Back course approach
CT	Contact approach
FA	Final approach
FMS	Flight management system approach
GPS	GPS approach
I	Initial approach
ILS	ILS approach
MA	Missed approach
MLS	MLS approach
NDB	Nondirectional radio beacon approach
OTP	VFR conditions-on-top
PA	Precision approach
PT	Procedure turn
RA	Resolution advisory (Pilot reported TCAS event)
RH	Runway heading
RNAV	Area navigation approach
RP	Report immediately upon passing (fix/altitude)
RX	Report crossing
SA	Surveillance approach
SI	Straight-in approach
TA	TACAN approach
TL	Turn left
TR	Turn right
VA	Visual approach
VR	VOR approach

APPENDIX B: CONTROL INFORMATION SYMBOLS TABLE

Control Information Symbols (Part I)

JO 7110.65,
par. 2-3-10,
figure 2-3-7

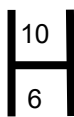
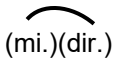
Symbols	Meaning
T → ()	Depart (direction, if specified)
↑	Climb and Maintain
↓	Descend and Maintain
→	Cruise
@	At
X	Cross
—M→	Maintain
≡	Join or intercept airway/jet route/track or course
≡	While in controlled airspace
△	While in control area
△	Enter control area
△	Out of control area
NW ↘ NE ↗ E →	Cleared to enter, depart or through surface area. Indicated direction of flight by arrow and appropriate compass letter. Maintain Special VFR conditions (altitude if appropriate) while in surface area.
250 K	Aircraft requested to adjust speed to 250 knots.
-20 K	Aircraft requested to reduce speed 20 knots
+30 K	Aircraft requested to increase speed 30 knots
W	Local Special VFR operations in the vicinity of (name) airport are authorized until (time). Maintain special VFR conditions (altitude if appropriate)
>	Before
<	After or past
<u>170</u> (red)	Inappropriate altitude for direction of flight. (Underline assigned altitude/flight level in red).
/	Until
()	Alternate instructions
Restriction	Restriction
↓	At or Below
↑	At or Above
- (Dash)	From-to (route, time, etc.)
(Alt)B(Alt)	Indicates a block altitude assignment. Altitudes are inclusive, and the first altitude shall be lower than the second. Ex: 310B370
V <	Clearance void if aircraft not off ground by (time)
	NOTE: The absence of an airway route number between two fixes in the route of flight indicates "direct"; no symbol or abbreviation is required.

Continued on next page

APPENDIX B: CONTROL INFORMATION SYMBOLS

TABLE (Continued)

**Control
Information
Symbols
(Part II)**
JO 7110.65,
par. 2-3-10,
figure 2-3-8

	Pilot cancelled flight plan
✓	EN ROUTE: Aircraft has reported at assigned altitude, Example; 80
✓	TERMINAL/FSS: Information forwarded (indicated information forwarded as required)
 (red)	EN ROUTE: Information or revised information forwarded. (altitude/flight level for direction of flight or other control information). Also circle, in red, the time (Minutes and altitude when a flight information forwarded. Use method in both inter-center and intra-center)
	Other than assigned altitude reported (circle reported altitude)
	DME holding (use with mileages)(Upper figure indicates distance from station to DME fix, lower figure indicates length of holding pattern.) In the example, the DME fix is 10 miles out with a 6 mile pattern indicated.
	DME arc of VORTAC, TACAN, or MLS.
C (freq.)	Contact (facility) or (Freq.), (time, fix, or altitude if appropriate). Insert frequency only when it is other than standard.
R	Radar contact
R	EN ROUTE: Requested altitude (preceding altitude information)
	Radar service terminated
	Radar contact lost
RV	Radar vector
	Pilot resume own navigation
	Radar handoff (circle symbol when handoff completed)
E (red)	EMERGENCY
W (red)	WARNING
P	Point out initiated. Indicate the appropriate facility, sector or position. Example: PZFW.
FUEL	Minimum fuel
	NOTE: The absence of an airway route number between two fixes in the route of flight indicates "direct"; no symbol or abbreviation is required.